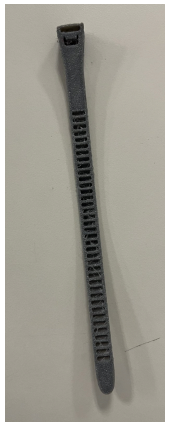
OUR PROTOTYPE

A splint made for one astronaut

Our value: OrthoPrint gives an injured astronaut a splint made for their own ankle, printed on the lunar base and strapped on with one hand.



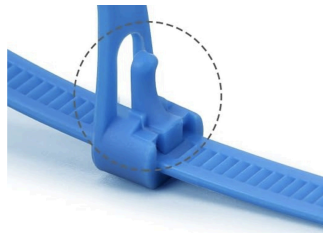
Strap, front



Side



Back

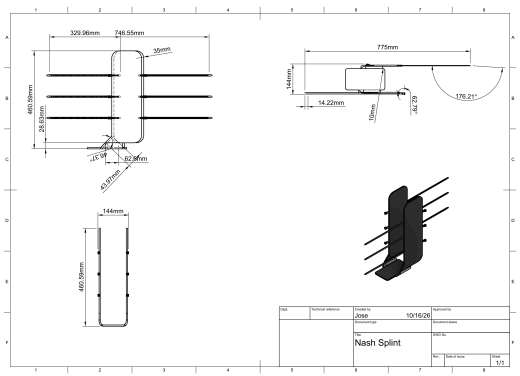


The reusable zip tie behind our closure

See the full CAD
gertrudish978-pixel.github.io/lunar-splint

OTHER DESIGN FEATURES

Details & next steps



Detail drawing, dimensions in mm · Jose

Materials

PLA · ABS · TPU · PETG

CF-PEEK for flight

Size	460.59 mm tall, side plates 144 mm apart
Fit	Unlatches and goes on from above
Shell	Solid TPU wall, firm but moulds slightly

Next steps

- ○ Test strap thickness and break load
- ○ Find where the filament softens with heat
- ○ Build prototype #2 for the CDR

Gladys Porter Early College High School
Brownsville, Texas

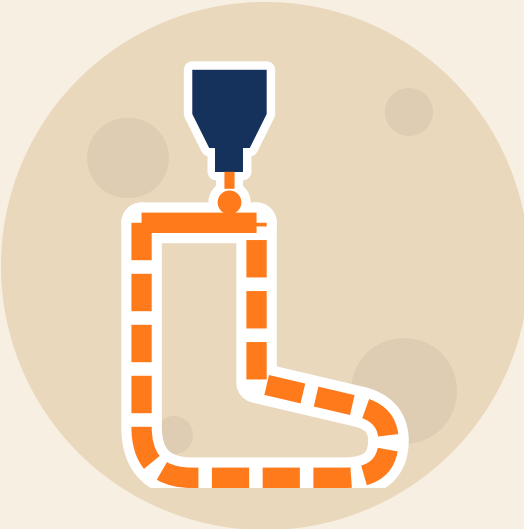
A student design project. Not a medical device.

LUNOVA

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

OrthoPrint

A custom-fit ankle splint, 3D-printed on the Moon.



Team Lunova · Gertrudis, Jose & Nancy
Gladys Porter Early College High School
Brownsville, Texas
Teachers: Lupita C. Vasquez and Claudia Cortez

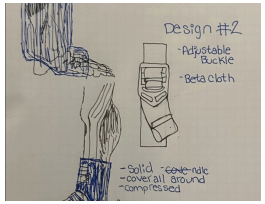
Design & Prototype · 2026–27

Website, CAD & videos
gertrudish978-pixel.github.io/lunar-splint

PROTOTYPE HIGHLIGHTS

Our timeline

- **Aug** — Formed the team, chose the challenge and ran our Needs Assessment Survey
- **Sept** — Research and brainstorming: a one-size cast became a two-piece splint
- **Sept 18** — Concept decision signed
- **Sept 24** — Solid prototype and a new latch design
- **Oct** — Prototype #1: printed and tested PETG, TPU and straps
- **Oct 16** — New CAD and detail drawing
- **Oct 21–22** — **Preliminary Design Review**
- **Nov–Dec** — Prototype #2 for the CDR



Early concepts



Holes split, so we went solid

What we tested

PETG with holes broke when bent	NO
TPU with holes layers split	NO
Solid TPU firm, moulds slightly	CHOSEN
Straps at key points ankle stayed still	CHOSEN

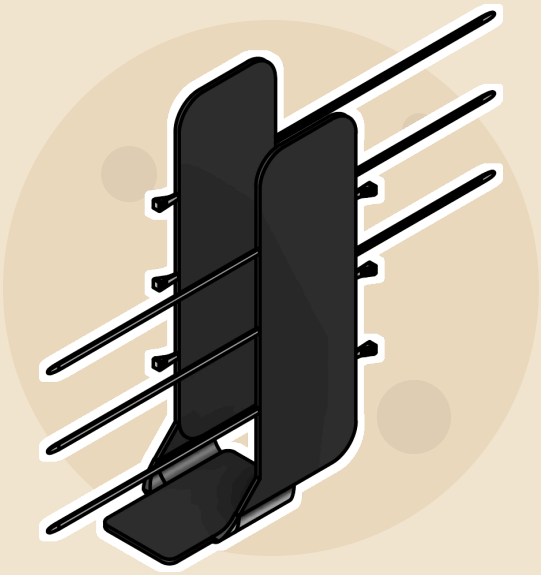
WHAT MAKES IT UNIQUE

Built for the Moon

1. **Single-piece ratchet closure**
   Tightens or releases with one hand. No loose parts to lose.

2. **Integrated hygiene**
   Comes off to clean the splint and check the skin.

3. **Thermal phase design**
   Flexible to put on, firm to hold the ankle still.



Our 3D CAD (isometric) · Jose

BEYOND THE REQUIREMENTS

More than the challenge asks

3	lunar innovations, where the challenge asks for one
1	splint per patient, custom-fit from a scan instead of fixed sizes
56	students in our Needs Assessment Survey, plus an athletic-trainer interview
4	print materials tested, with CF-PEEK picked for flight

What our survey told us

Ankle injured most	63%
Hard to find a fit	54%
On their feet a lot	96%
Devices help healing	98%

Why the Moon needs it

$1,000 to launch every gram, so no box of spare sizes

Days for a ride home, so the crew makes the splint on site